

A SAT-Modulo-Symmetries verification of the Erdős–Gyárfás power-of-two cycle conjecture for minimum-degree-3 graphs up to 30 vertices

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Abstract

The Erdős–Gyárfás conjecture (1995) asserts that every graph with minimum degree at least 3 contains a cycle whose length is a power of two. The conjecture is open; prior computer search established that any counterexample on the *general* minimum-degree-3 class has at least 17 vertices (Royle and Markström), and that any *cubic* (3-regular) counterexample has at least 30 vertices (Markström, 2004). We give, to our knowledge, the first application of SAT-based methods to this conjecture. Using SAT Modulo Symmetries (SMS) with the Glasgow subgraph solver as a complete forbidden-subgraph propagator, we verify that *every* minimum-degree-3 graph on at most 30 vertices contains a cycle of length 4, 8, or 16. Consequently any general minimum-degree-3 counterexample must have at least 31 vertices, improving the two-decade-old general bound from 17 to 31; since the cubic class is a subclass, this also improves the cubic bound from 30 to 31. We corroborate the result with an independent ground-truth check at $n = 10$, reproduction of the $n \leq 16$ baseline, agreement with an independent CEGAR-SAT solver for $n \leq 19$, robustness to two cardinality encodings and a second symmetry-breaking method, and positive controls; we discuss the path to a formally machine-checked proof certificate.

1 Introduction

In 1995 Erdős and Gyárfás posed the following conjecture (see Erdős [1]).

Conjecture 1 (Erdős–Gyárfás). *Every graph with minimum degree at least 3 contains a simple cycle whose length is a power of two.*

Here “power of two” means 2^k for some integer $k \geq 2$, i.e. a cycle of length 4, 8, 16, The conjecture remains open; Erdős offered \$100 for a proof and \$50 for a counterexample [2]. It has been confirmed for several restricted classes, including 3-connected cubic planar graphs [4], P_8 -free graphs [5], and P_{10} -free graphs [6]; the most recent structural progress shows that any *minimal* counterexample is “predominantly cubic” [7].

On the computational side, two distinct frontiers are known, and the distinction is central to the present paper. Write $\delta(G)$ for the minimum degree of G .

- **General** ($\delta \geq 3$, arbitrary maximum degree): computer searches of Royle and Markström established that any counterexample has at least 17 vertices, i.e. Conjecture 1 holds for all $\delta \geq 3$ graphs on at most 16 vertices [12, 3].

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- **Cubic** (3-regular): Markström [3] showed that any cubic counterexample has at least 30 vertices (all cubic graphs on at most 29 vertices were checked).

To our knowledge these bounds have not been improved since 2004 [TO BE CONFIRMED BY LITERATURE REVIEW].

Contribution. We give the first SAT-based attack on Conjecture 1. Using SAT Modulo Symmetries (SMS) [8, 9] together with the Glasgow subgraph solver [10], we verify Conjecture 1 for *all* minimum-degree-3 graphs on n vertices for every n up to 30:

Theorem 1. *Every graph with minimum degree at least 3 on at most 30 vertices contains a cycle of length 4, 8, or 16. Consequently, any minimum-degree-3 counterexample to the Erdős–Gyárfás conjecture has at least 31 vertices.*

This improves the general bound from 17 to 31. Because the cubic class is contained in the minimum-degree-3 class, Theorem 1 also verifies all cubic graphs on at most 30 vertices, improving the cubic bound from 30 to 31 as well.

We emphasize what is and is not new. The genuinely new content is the *non-cubic* minimum-degree-3 graphs on $17 \leq n \leq 30$ vertices (the cubic subcase on ≤ 29 vertices was already settled in [3]), and the methodology (SAT/SMS), which had not previously been applied to this conjecture.

2 Method

For a fixed number of vertices n we decide the following question with a single SMS call: *does there exist a graph G on n vertices with $\delta(G) \geq 3$ that contains no cycle of length 4, 8, or 16?* If the answer is “no” for every $n \leq 30$, then no minimum-degree-3 counterexample on ≤ 30 vertices exists, establishing Theorem 1. (For $n \leq 30$ the only power-of-two cycle lengths that can occur are 4, 8, 16, since $32 > 30$; thus forbidding C_4, C_8, C_{16} as subgraphs is exactly forbidding all power-of-two cycles.)

SAT Modulo Symmetries. SMS performs complete, isomorph-free generation of graphs satisfying given constraints: its modified CaDiCaL solver carries an in-search canonicity (minimality) propagator that prunes any partial adjacency matrix that cannot be the lexicographically minimal representative of its isomorphism class [8]. This is the key to scaling: the search considers essentially one graph per isomorphism class rather than enumerating all labelings.

Encoding. The minimum-degree constraint $\delta(G) \geq 3$ is encoded as a CNF cardinality constraint over the $\binom{n}{2}$ edge variables (we use SMS’s built-in sequential-counter encoding; see §4 for a totalizer cross-check). The constraint “no C_4, C_8 , or C_{16} ” is enforced by the Glasgow subgraph solver [10] as a complete forbidden-subgraph propagator: whenever a (partial) graph contains one of these cycles as a (non-induced) subgraph, the propagator adds a clause excluding it. Forbidding the short cycle C_4 statically in CNF is cheap, but C_8 and C_{16} have $\Theta(n^8)$ and $\Theta(n^{16})$ potential occurrences, so a static encoding is infeasible; the propagator is what makes the search tractable.

Why not plain CEGAR-SAT. We also implemented a self-contained counterexample-guided (CEGAR) SAT solver in Python (PySAT/CaDiCaL, with a DFS power-of-two cycle detector and a lexicographic symmetry break). With only a partial symmetry break it re-discovers each bad cycle in many isomorphic copies; it reaches $n = 19$ (bound ≥ 20) before the per-instance time grows out

of reach. SMS’s complete symmetry breaking is precisely what removes this bottleneck. The two solvers agree on $n \leq 19$ (§4).

3 Results

Table 1 reports, for each n from 17 to 30, that SMS finds *no* minimum-degree-3 graph avoiding all of C_4, C_8, C_{16} (solution count 0). All runs used a single CPU core.

n	count	wall time (s)
17	0	2.9
18	0	7.8
19	0	24.1
20	0	27.8
21	0	19.3
22	0	101.1
23	0	202.9
24	0	148.3
25	0	339.8
26	0	414.8
27	0	1000.3
28	0	1892.0
29	0	2342.8
30	0	6888.9

Table 1: SMS decision per order n : no minimum-degree-3 graph on n vertices avoids C_4, C_8, C_{16} . Together with the reproduced $n \leq 16$ baseline this proves Theorem 1.

4 Verification

A non-existence claim at $n = 30$ cannot be brute-forced (exhaustive generation of minimum-degree-3 graphs is infeasible beyond roughly $n = 13$), so we corroborate Theorem 1 with several independent checks. *No check produced a contradiction.*

1. **Ground-truth anchor.** At $n = 10$, forbidding only C_4 , SMS returns exactly 5 graphs, matching the independent count of the 5 C_4 -free minimum-degree-3 graphs on 10 vertices obtained with `nauty` (`geng+labelg`) [11].
2. **Baseline reproduction.** For $n = 6, \dots, 16$, forbidding all of C_4, C_8, C_{16} yields 0, reproducing the published baseline.
3. **Independent second solver.** Our CEGAR-SAT solver and SMS both return UNSAT for $n = 17, 18, 19$ (different solvers, symmetry breaking, and cycle handling).
4. **Encoding robustness.** Re-deciding with the totalizer cardinality encoding (a structurally different CNF) gives count 0 at $n = 17, 20, 22, 25$.
5. **Symmetry-method robustness.** Re-deciding with SMS’s colex minimality ordering gives count 0 at $n = 17, 20$; the (slower, experimental) colex variant timed out at $n = 22, 25$ (inconclusive, not contradictory).

6. **Positive controls.** Forbidding only C_4 (a weaker constraint) yields a graph at $n = 17, 20, 25, 30$, confirming the pipeline returns a solution when one exists.

Toward a machine-checked certificate. `smsg` can emit an LRAT proof for UNSAT instances, but a generic LRAT checker cannot validate it against the minimum-degree-3 CNF alone, because the forbidden-cycle clauses are added by the propagator during search and are not RUP/RAT-derivable from that CNF (they exclude valid minimum-degree-3 graphs). A fully machine-checked certificate therefore requires the certified-SMS proof-logging machinery, in which the propagator justifies its own clauses; we leave this as the principal next step.

5 Discussion and limitations

Theorem 1 advances the general minimum-degree-3 frontier from 16 to 30 vertices (bound $17 \rightarrow 31$) and, as a byproduct, the cubic frontier from 29 to 30 (bound $30 \rightarrow 31$). We stress two caveats. First, this is a *computational* result whose soundness rests on SMS’s isomorph-free generation, the completeness of the Glasgow subgraph propagator, and the correctness of the min-degree encoding; the checks in §4 validate this composition at $n = 10$ and across $n \leq 16$, but a third-party independent reproduction and/or a formal proof certificate would be needed for a fully rigorous claim. Second, the result does not prove the conjecture; it only raises the lower bound on the size of a hypothetical counterexample.

Reproducibility

All code, the per- n data, and the verification scripts are available at <https://github.com/ArjunBalaji79/erdos-gyarfas-min-degree-3>.

Acknowledgments

The computational pipeline was developed and executed with substantial AI assistance; all claims are subject to the independent-reproduction caveat above.

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